MCQs – SEARCHING (Computer Science, Class XII)

INTRODUCTION TO SEARCHING

1. Searching in computer science refers to
   A) Sorting elements in a list
   B) Locating a particular element in a collection
   C) Deleting elements from a list
   D) Rearranging elements in memory

2. The result of a search operation determines
   A) The size of the list
   B) The type of data stored
   C) Whether an element is present or not
   D) The sorting order of the list

3. If a searched element is found, searching can also determine
   A) The frequency of the element
   B) The memory size used
   C) The position of the element
   D) The data type of the element

4. Searching is considered an important technique in computer science because
   A) It reduces memory usage
   B) It helps in designing algorithms
   C) It avoids data duplication
   D) It encrypts data

5. A computer stores data mainly to
   A) Display graphics
   B) Retrieve it later when demanded
   C) Compress files
   D) Format storage devices

LINEAR SEARCH – CONCEPT

6. Linear search is also known as
   A) Binary search
   B) Random search
   C) Sequential search
   D) Indexed search

7. Linear search compares elements
   A) Randomly
   B) From last to first
   C) One by one
   D) Only at fixed positions

8. In linear search, comparison starts from
   A) Middle element
   B) Last element
   C) First element
   D) Any element

9. Linear search is an example of
   A) Divide and conquer technique
   B) Exhaustive searching technique
   C) Hashing technique
   D) Recursive technique

10. Linear search is most useful when the list is
    A) Large and sorted
    B) Small and unordered
    C) Large and unordered
    D) Small and sorted

11. If no element matches the key in linear search, the search is declared
    A) Partially successful
    B) Deferred
    C) Unsuccessful
    D) Repeated

LINEAR SEARCH – ALGORITHM & WORKING

12. In Algorithm 6.1, the variable index is initially set to
    A) 1
    B) n
    C) 0
    D) −1

13. The loop in linear search continues while
    A) index > n
    B) index == n
    C) index < n
    D) index ≤ n−1 only

14. In linear search, the comparison performed is
    A) numList[index] > key
    B) numList[index] < key
    C) numList[index] = key
    D) numList[index] ≠ key

15. When a key is found, the algorithm prints
    A) The index value
    B) The element value
    C) The position as index + 1
    D) The number of comparisons

16. If the key is not found after traversing the list, the algorithm prints
    A) Element absent
    B) Search terminated
    C) Search unsuccessful
    D) Key invalid

LINEAR SEARCH – EXAMPLES & OBSERVATIONS

17. If the key is the first element of the list, linear search requires
    A) n comparisons
    B) n−1 comparisons
    C) 1 comparison
    D) 2 comparisons

18. If the key is the last element in the list, linear search requires
    A) n comparisons
    B) n−1 comparisons
    C) 1 comparison
    D) n/2 comparisons

19. When the key is not present in the list, linear search performs
    A) One comparison
    B) n/2 comparisons
    C) n comparisons
    D) Zero comparisons

20. The maximum work done by linear search occurs when
    A) Key is the first element
    B) Key is the middle element
    C) Key is the last element
    D) Key is not present

21. The minimum work done by linear search occurs when
    A) Key is absent
    B) Key is the last element
    C) Key is the first element
    D) List is empty

LINEAR SEARCH – PROGRAM BASED

22. The linear search function returns
    A) Key value
    B) Index position
    C) Index + 1 or None
    D) Boolean value

23. If the function returns None, it means
    A) The list is empty
    B) The key is not present

C) The key is present multiple times
D) The list is sorted

24. In the program, user input is taken to
    A) Sort the list
    B) Create the list
    C) Delete elements
    D) Reverse the list

25. The list used in the program is initially
    A) Predefined
    B) Sorted
    C) Empty
    D) Random

BINARY SEARCH – CONCEPT

26. Binary search works efficiently only when the list is
    A) Unordered
    B) Random
    C) Sorted
    D) Reversed

27. Binary search makes use of
    A) Exhaustive checking
    B) Hash function
    C) Ordering of elements
    D) Stack memory

28. Binary search compares the key with
    A) First element
    B) Last element
    C) Middle element
    D) Random element

29. Binary search reduces the search space by
    A) One element
    B) Two elements
    C) Half
    D) One-third

30. Binary search is inspired by searching words in
    A) Newspaper
    B) Telephone directory
    C) Dictionary
    D) Encyclopedia

BINARY SEARCH – POSSIBLE CASES

31. If the middle element matches the key, the search is
    A) Repeated
    B) Terminated successfully
    C) Continued
    D) Restarted

32. If the middle element is greater than the key, the search continues in
    A) Entire list
    B) Second half
    C) First half
    D) Random half

33. If the middle element is smaller than the key, the search continues in
    A) First half
    B) Second half
    C) Entire list
    D) Middle portion

BINARY SEARCH – MID CALCULATION

34. The middle position is calculated using
    A) Normal division
    B) Modulo operator
    C) Floor division
    D) Ceiling function

35. If a list has 10 elements, the middle index is
    A) 4
    B) 5
    C) 6
    D) 10

36. The first element of a list has index value
    A) −1
    B) 0
    C) 1
    D) Depends on list size

BINARY SEARCH – ITERATIONS & PERFORMANCE

37. Binary search uses the term iteration because
    A) It uses recursion
    B) It avoids comparison
    C) Search area is redefined each time
    D) It sorts the list repeatedly

38. Each unsuccessful comparison in binary search
    A) Increases search space
    B) Leaves search space unchanged

C) Reduces search space by half
D) Doubles comparisons

39. Binary search requires minimum work when the key is
    A) First element
    B) Last element
    C) Middle element
    D) Absent

40. Binary search requires maximum work when
    A) Key is the middle element
    B) Key is the first element
    C) Key is absent or near extremes
    D) List is empty

BINARY SEARCH – ALGORITHM & PROGRAM

41. In binary search algorithm, initial values are
    A) first = 1, last = n
    B) first = 0, last = n−1
    C) first = mid
    D) last = 0

42. The binary search loop runs while
    A) first < last
    B) first == last
    C) first <= last
    D) mid <= last

43. If key is greater than middle element,
    A) last is decreased
    B) first is increased
    C) mid is unchanged
    D) search stops

44. If key is smaller than middle element,
    A) first = mid + 1
    B) last = mid − 1
    C) mid = 0
    D) search stops

45. The binary search function returns −1 when
    A) Key is found
    B) List is empty
    C) Key is not found
    D) List is unsorted

APPLICATIONS OF BINARY SEARCH

46. Binary search can be used for
    A) Searching a dictionary
    B) Telephone directory
    C) Finding minimum/maximum in sorted list
    D) All of the above

47. Modified binary search techniques are used in
    A) Image editing
    B) Database indexing
    C) File compression only
    D) Email services

SEARCH BY HASHING – CONCEPT

48. Hashing allows searching in
    A) Linear time
    B) Logarithmic time
    C) One step
    D) Multiple passes

49. Hashing uses a
    A) Sorting algorithm
    B) Hash function
    C) Binary tree
    D) Stack

50. A hash function generates
    A) Data type
    B) Index value
    C) Sorted list
    D) Key

51. The new list created using hashing is called
    A) Search list
    B) Sorted list
    C) Hash table
    D) Linked list

HASH TABLE & HASH FUNCTION

52. Each index of a hash table can hold
    A) Multiple items
    B) Only one item
    C) No items
    D) Two items

53. Hash table indices start from
    A) 1
    B) −1

C) 0
D) Depends on size

54. The size of hash table can be
    A) Equal to list size only
    B) Smaller than list size only
    C) Larger than list size
    D) Always fixed

55. The remainder method hash function is
    A) h(key) = key // size
    B) h(key) = key * size
    C) h(key) = key % size
    D) h(key) = size % key

HASHING – SEARCH & PERFORMANCE

56. Hash-based searching requires
    A) Multiple comparisons
    B) n comparisons
    C) One comparison
    D) log n comparisons

57. Hashing performance is independent of
    A) Hash function
    B) List size
    C) Table size
    D) Key value

COLLISION & PERFECT HASHING

58. Collision occurs when
    A) Key is not found
    B) Two elements map to same index
    C) Hash table is full
    D) Key is repeated

59. Collision resolution refers to
    A) Deleting keys
    B) Rehashing table
    C) Placing items with same hash value
    D) Sorting hash table

60. A perfect hash function ensures
    A) Fast sorting
    B) Zero collisions
    C) Multiple keys per index
    D) Larger hash table

SUMMARY & CONCEPTUAL MCQs

61. Linear search checks elements
    A) Randomly
    B) One at a time
    C) By dividing list
    D) Using hash function

62. Binary search is faster than linear search because
    A) It skips comparisons
    B) It halves the search space
    C) It uses recursion
    D) It uses hashing

63. Hashing is efficient because
    A) It sorts data
    B) It reduces memory
    C) Index computation time is constant
    D) It avoids collisions

64. When two elements map to the same slot, it is called
    A) Iteration
    B) Comparison
    C) Collision
    D) Overflow

65. Collision resolution techniques are
    A) Discussed in detail
    B) Not required
    C) Beyond scope of the book
    D) Mandatory for binary search